FABEHA FATIMA

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Education

University of Massachusetts Amherst Amherst, USA
Master of Science in Computer Science (conc. Data Science); GPA: 3.9/4.0 *September 2024 – May 2026*

Amity University Noida, India
Bachelor of Technology in Computer Science and Engineering; GPA: 3.9/4.0 *July 2017 – July 2021*

Relevant Coursework: Distributed & Operating Systems, Advanced Algorithms, Theory & Practice of Software Engineering, Systems for Data Science, Data Structures using C, Object Oriented Programming using C++

Work Experience

Graphite San Francisco, CA (Remote)
Graduate Research Extern (AI Engineer) *January 2026 – May 2026*

- Researched how AI-generated content poisons the retrieval systems built on top of it, simulating collapse across *4* LLM families with **vLLM**, **FAISS**, and **SLURM**; found entity diversity dropped *73%* and lexical diversity *45%*.
- Benchmarked paraphrasing, prompt optimization, re-ranking, and agentic retrieval as collapse mitigations via **LLM** evaluation.
- Deployed agentic retrieval strategy as a production **RAG** system that autonomously calls tools and retrieves information under built-in guardrails; cut failure rate to *27.2%* vs. *48–67%* for simpler baselines, a *4x* improvement.
- Co-authored ***"Does RAG Collapse When It Retrieves AI-Generated Documents?"***, submitted to EMNLP 2026 workshop.

Cisco Systems Bengaluru, India
Software Engineer II *November 2022 – August 2024*

- Engineered an observability platform with **ELK stack**, **Redis** caching, **CDN**-accelerated static asset delivery, and **Jenkins** CI/CD, serving **525 engineering teams** processing **1.09B+** daily data points; participated in on-call rotations to triage, debug, and resolve production incidents.
- Architected a multi-cloud containerized Zabbix infrastructure monitoring platform (**Linux**, **Docker**) with three-tier Nginx **load balancing** regional stacks, ELK clusters, and database using round-robin routing and primary-backup failover; migrated **35K+** devices across 3 regional stacks with automated health monitoring and **failover** under **15 seconds**, achieving **100%** resiliency and **$3.4M** annual cost savings.
- Developed a topology-aware placement engine in **Python** that onboarded **35K+** servers to optimal regional stacks via **IP geolocation** and hierarchical cluster affinity; implemented as an event-driven control loop with automated failover routing.
- Led a team of 5 engineers to develop and deploy a **RESTful Java Spring MVC** platform automating **8+** AppDynamics (**OpenTelemetry**) provisioning workflows for **4K+** applications; conducted code reviews enforcing Java coding standards, unit and integration testing practices, and software architecture; open-sourced on ***Cisco Code Exchange***.

Software Engineer I *January 2021 – October 2022*

- Designed and implemented a high-throughput notification system (**MySQL**-backed Java microservices and React frontend) dispatching **20K+** emails/day, authoring the full database schema and data model to support role-based alerting; improved migration coordination for **25K+** servers and drove cross-functional adoption across teams.
- Streamlined real-time observability across **525** engineering teams by building an event-driven **Logstash** pipeline (**Ruby**) to deduplicate and filter event streams before publishing to **Kafka**, reducing alert fatigue by **59%**.
- Designed, deployed, and maintained a **Spring Boot** self-service platform backed by **MongoDB** increasing application uptime to **99.9%** by reducing Mean Time To Detect (MTTD) and Mean Time To Repair (MTTR) across production services.

Projects

Query–Product Relevance & Sponsored Search *(GitHub)* *April 2026 – Present*

- Built a hybrid **BM25** and dense retrieval pipeline (Sentence-BERT, **FAISS ANN**) with cross-encoder re-ranking and **LambdaRank** LTR (**LightGBM**) over **1M+** query-product pairs; achieved **+14% nDCG@10**, **+11% MRR** over baseline.
- Engineered **Apache Spark** distributed ingestion and deduplication at scale with hard negative sampling; tracked **5** benchmark stages across Precision@K, MRR, and nDCG in **MLflow**.

TradeNet: Distributed Stock Exchange *(GitHub)* *February 2025 – May 2025*

- Engineered a real-time, fault-tolerant distributed trading platform on **AWS EC2** using Dockerized microservices with **REST** and **gRPC**; configured Nginx round-robin load balancing with health-check-driven failover, reducing latency by **67%** while sustaining **1K+** requests/sec at sub-100ms with automated failover under **10ms**.
- Implemented distributed systems design patterns including consensus-based leader election, state replication, and orchestration-driven failover; deployed health-check-driven recovery and horizontal autoscaling to sustain fault tolerance.

Technical Skills

Backend & Databases: Java, Python, PHP, Ruby, SQL, Bash, Spring Boot, gRPC, PostgreSQL, MySQL, MongoDB, Redis
Systems: Apache Kafka, Docker, Kubernetes, Linux, PySpark, Nginx, TCP/IP, CDN, AWS (EC2, S3), Terraform
Cloud & DevOps: Jenkins, CI/CD, Multi-cloud Deployments, Grafana, ELK Stack, Zabbix, Prometheus, OpenTelemetry
Engineering Practices: Testing, Test Automation, Code Reviews, Design Reviews, Agile/Scrum
AI/ML: vLLM, SLURM, PyTorch, TensorFlow, Hugging Face Transformers, RAG, FAISS, LangChain, LangGraph, Agent Orchestration, Prompt Engineering